



FINAL REVIEW

Sensi reads a floor plan and tells you how each room will feel, across six senses, for one person.

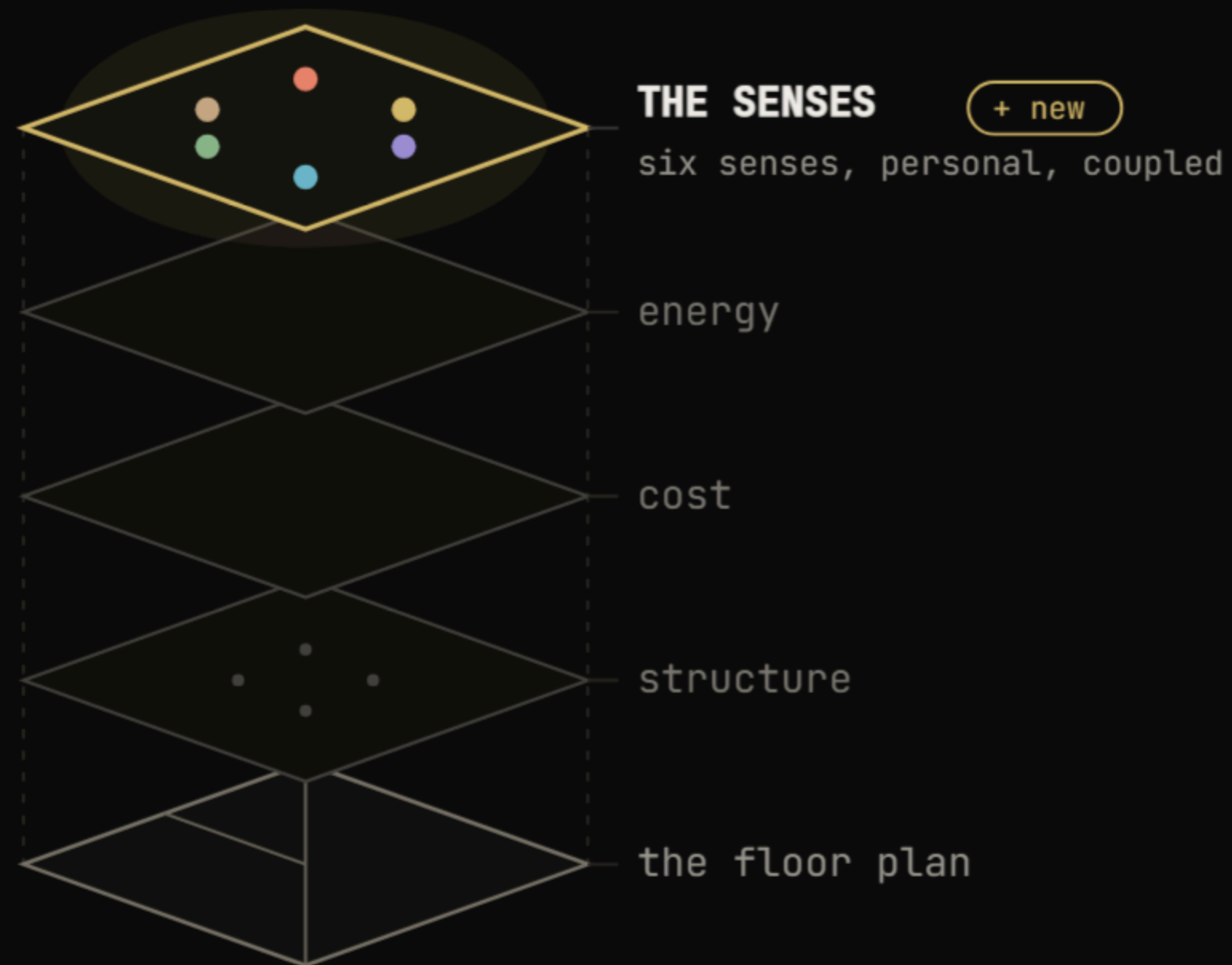
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— WHAT IS SENSI

A design copilot that adds the sensory layer.

how each room will feel, across six senses, while it is still a plan.



THE SENSORY LAYER IS

- **personal**
scored for one person, never an average
- △ **honest**
no averaging; the worst sense carries the weight
- ~ **coupled**
the senses pull on each other

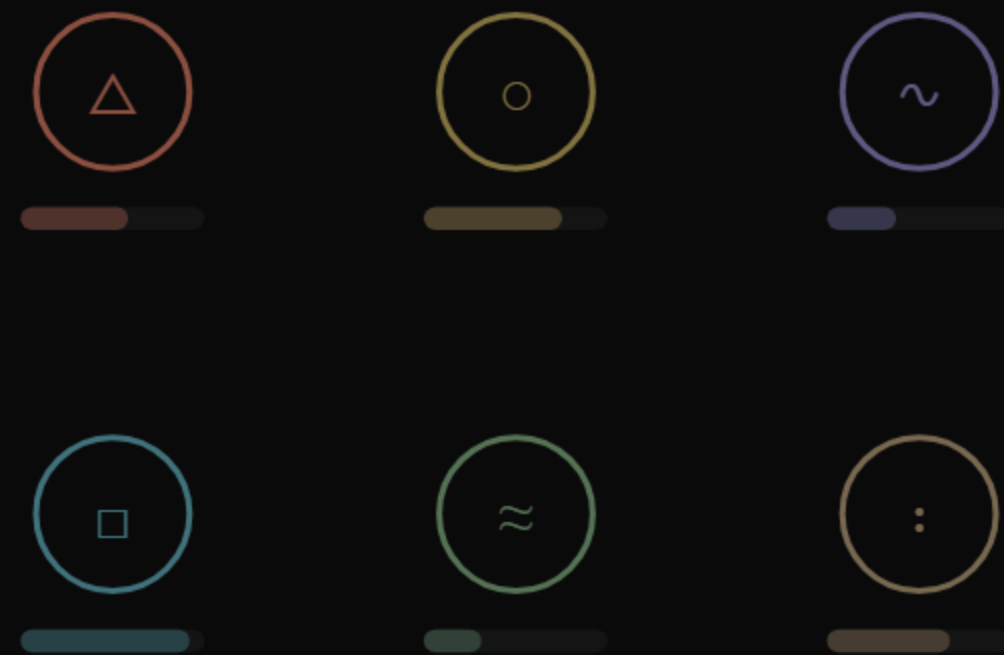
structure, cost, energy, code, and now: how a space feels.

— THE GAP IN THE AEC

Can we measure how a building will feel?

we measure energy, cost, code, but not how a room **feels**.

TODAY • MEASURED IN ISOLATION

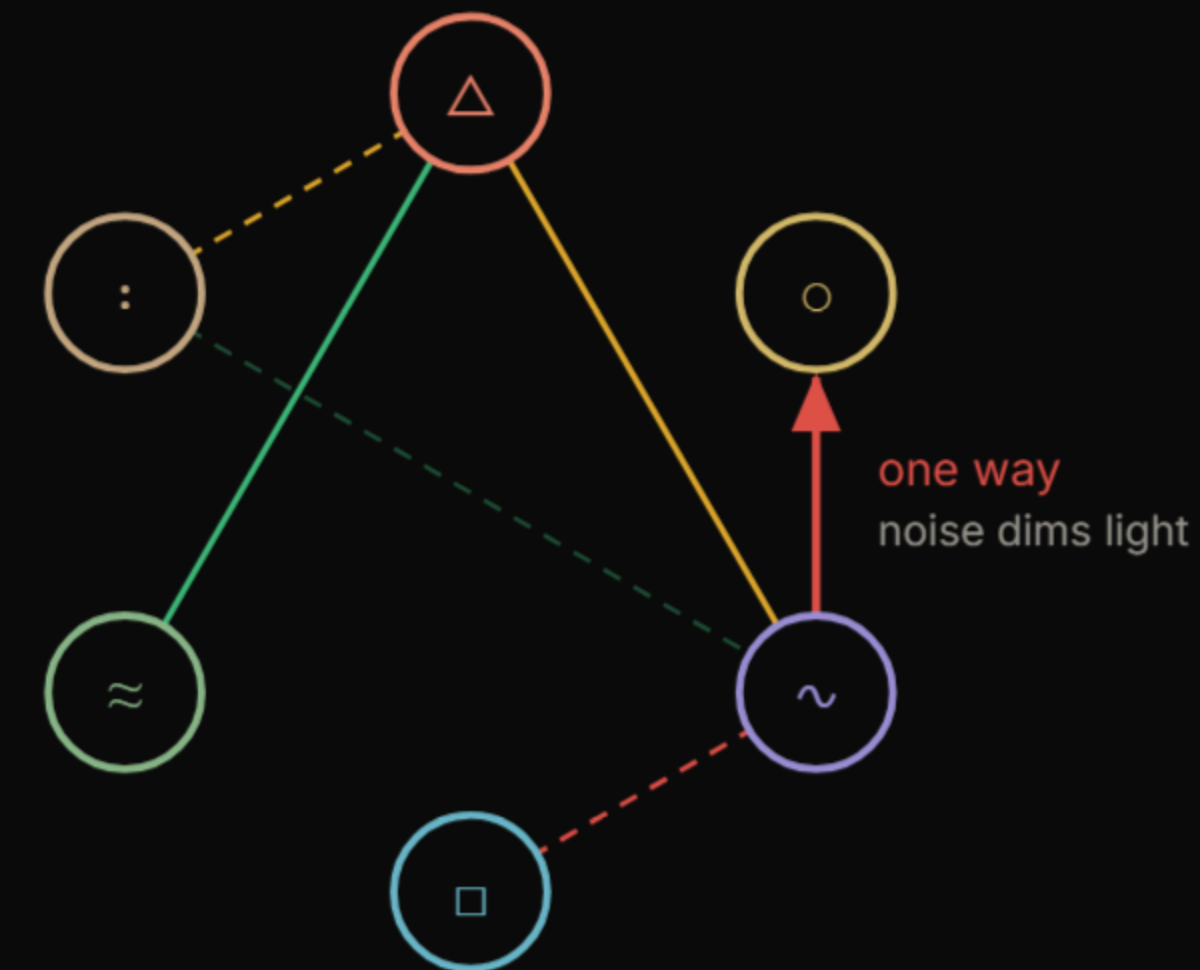


six separate gauges, one domain at a time

what's
missing



SENSI • ONE COUPLED SYSTEM



change one, the others move; some links run one way

R1 • in isolation

Al-Harasis, Jabi & Sharmin, 2025

a taxonomy for sensory design in autism: senses can be designed for

R2 • the bridge

Spence et al., 2020

senses of place, the multisensory mind: you take a room in all at once

R3 • coupled

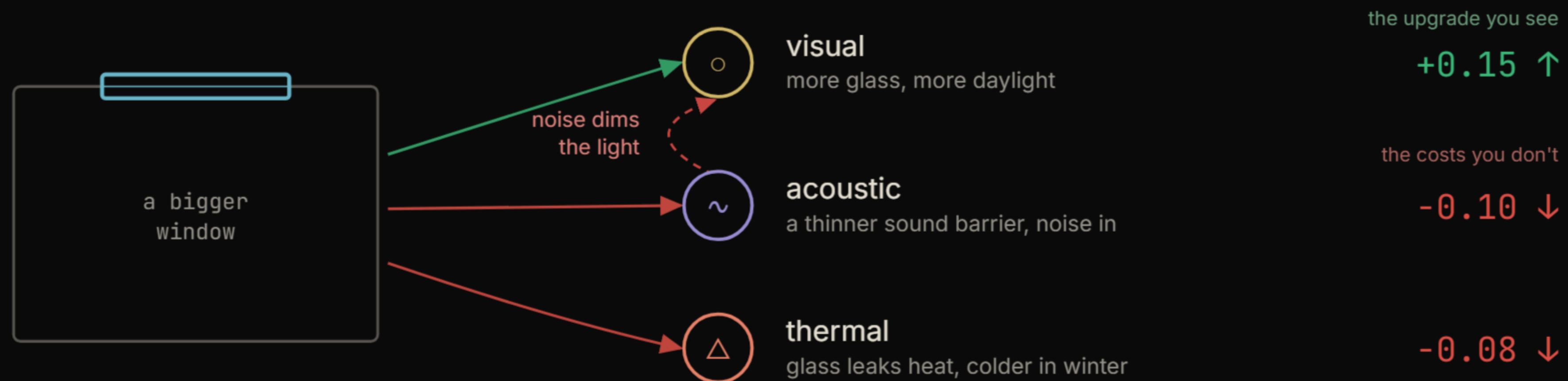
Multi-sensory thermal comfort review, 2025

cross-modal: one sense moderates another

THE RIPPLE

One move ripples through every sense.

Each standard scores a sense in its own silo, so the second cost, and the way it feeds back on the first, stays invisible until someone lives with it. **Could a designer catch this while it is still a plan?**

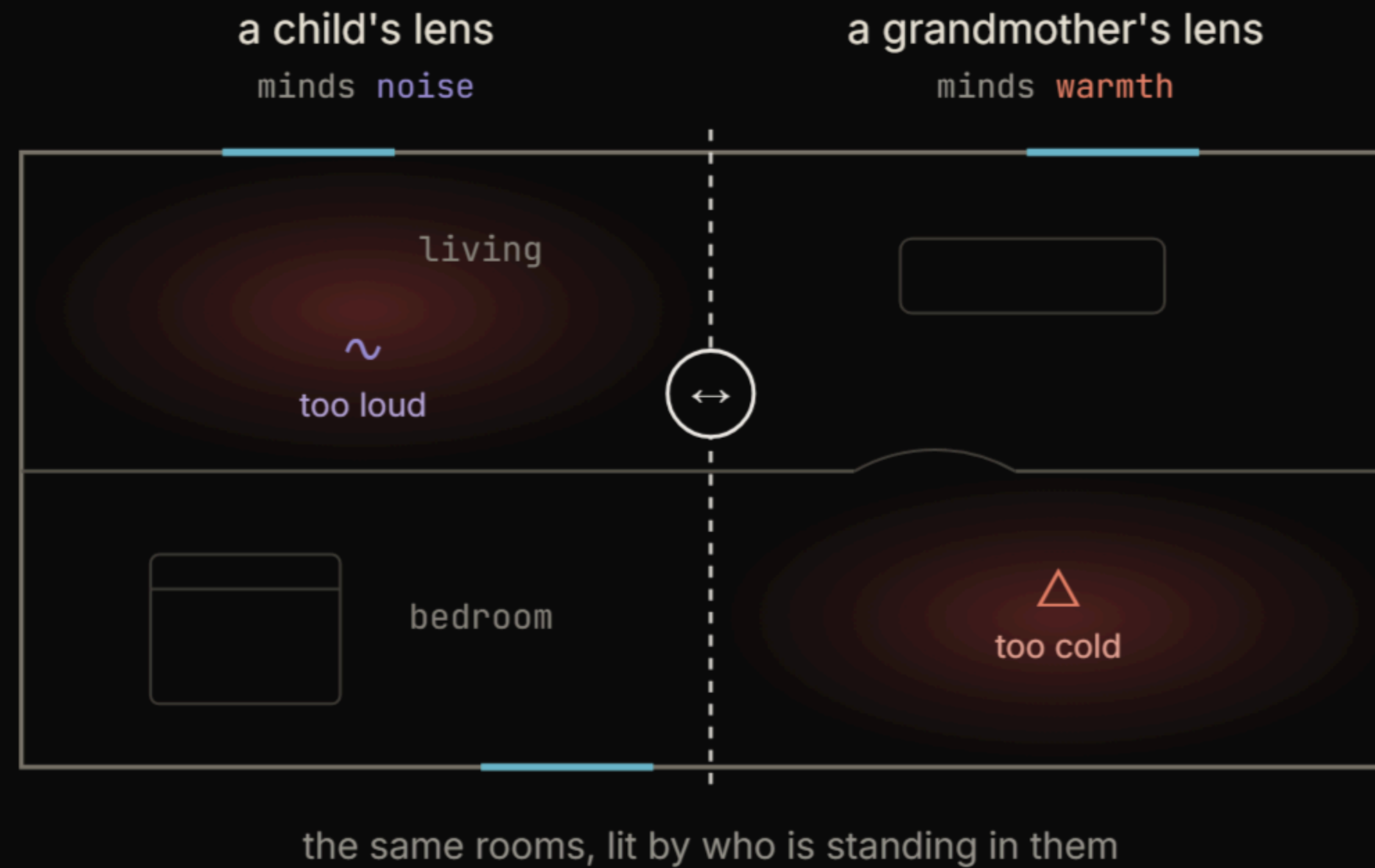


R3 • **coupled** Multi-sensory thermal comfort review, 2025 cross-modal: one sense moderates another, additive, synergistic, or compensatory

— THE SAME PLAN, TWO PEOPLE

The same plan feels different to different people.

comfort is personal: one plan, two lenses, and a different room lights up for each.



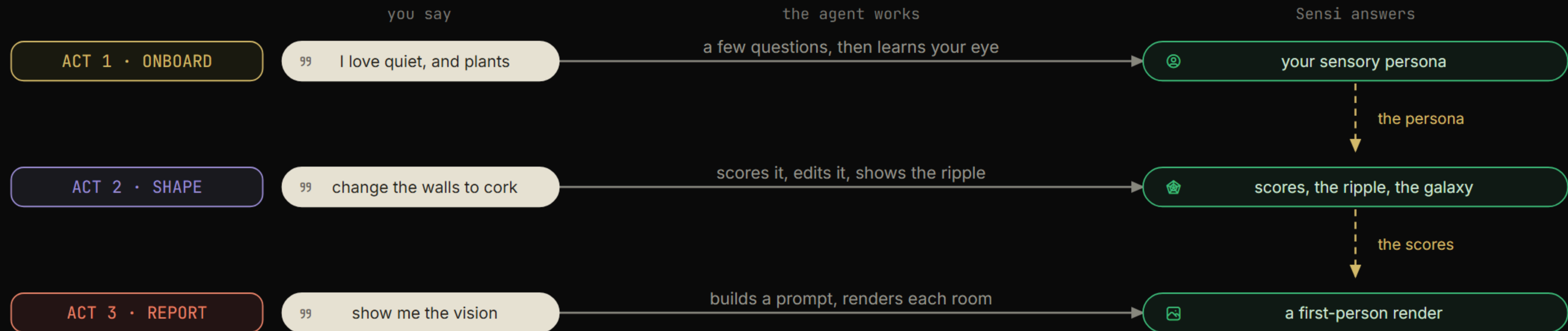
S4 • weights **Frontiers Built Env., 2026** comfort weights are context-dependent, not constants

R1 • sensitivity **Al-Harasis et al., 2025** sensory sensitivity is individual

— THE THREE ACTS

The whole thing is a conversation, in three acts.

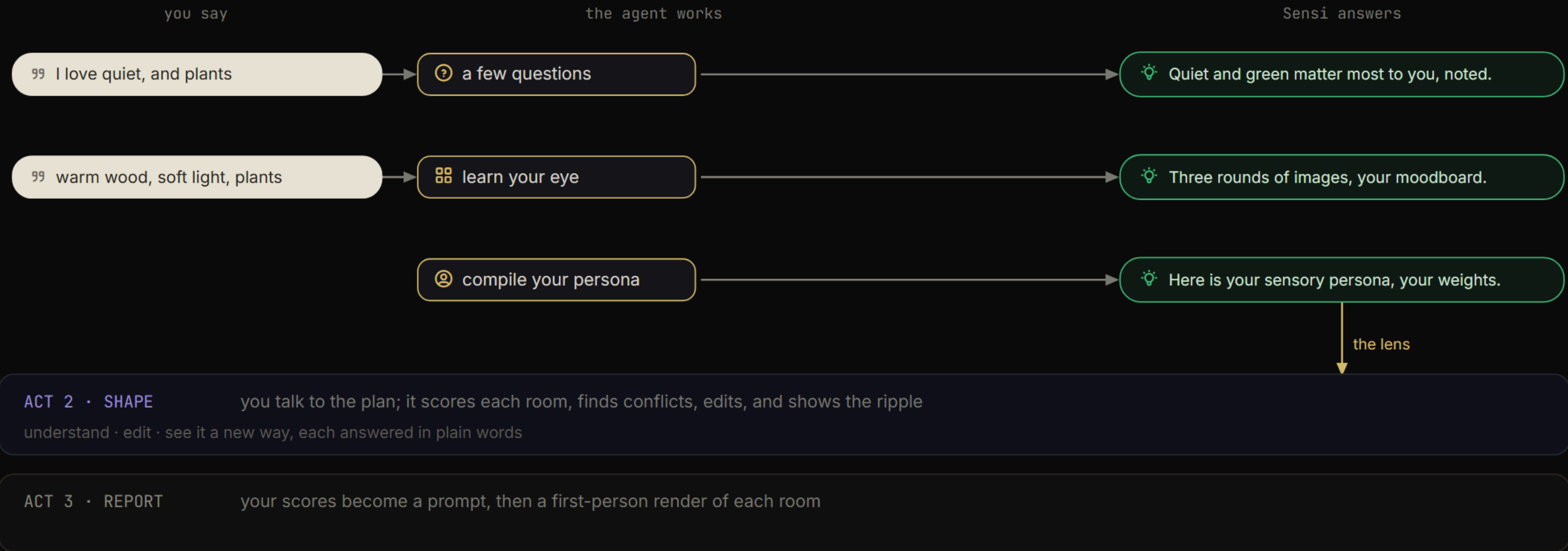
you talk, Sensi answers, and the output of each act carries into the next.



— ACT 1 / ONBOARD

Act one: Sensi gets to know you.

you answer a few questions and curate a moodboard; it compiles your sensory persona.



the persona you build here is the lens for everything that follows.



SENSI

| ONBOARDING

• 3 OF 7 • YOUR SPACE

Hi, I'm Sensi — who are you?

hello i am em

Nice to meet you, Em! How would you describe your relationship with spaces? Do you design them, live in them, or are you here to learn?

I design spaces -- architect



Design perspective — exactly what Sensi needs, Em. Tell me about a space that made you feel truly at home -- it could be anywhere. What made it feel right?

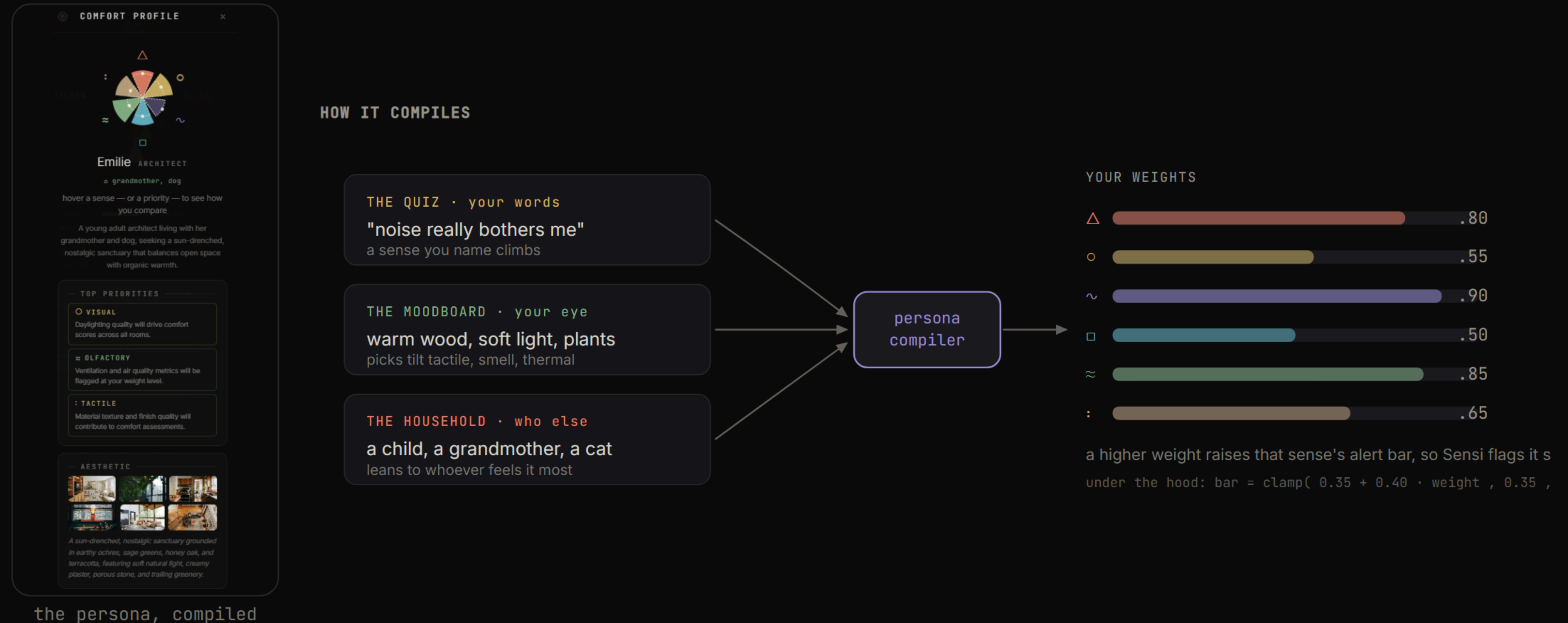
the kitchen of my friend, it was a large open space with window



IT'S PERSONAL

Three inputs compile into your persona

nothing you do is wasted: your words, your eye, and your household all become numbers, one weight per sense.

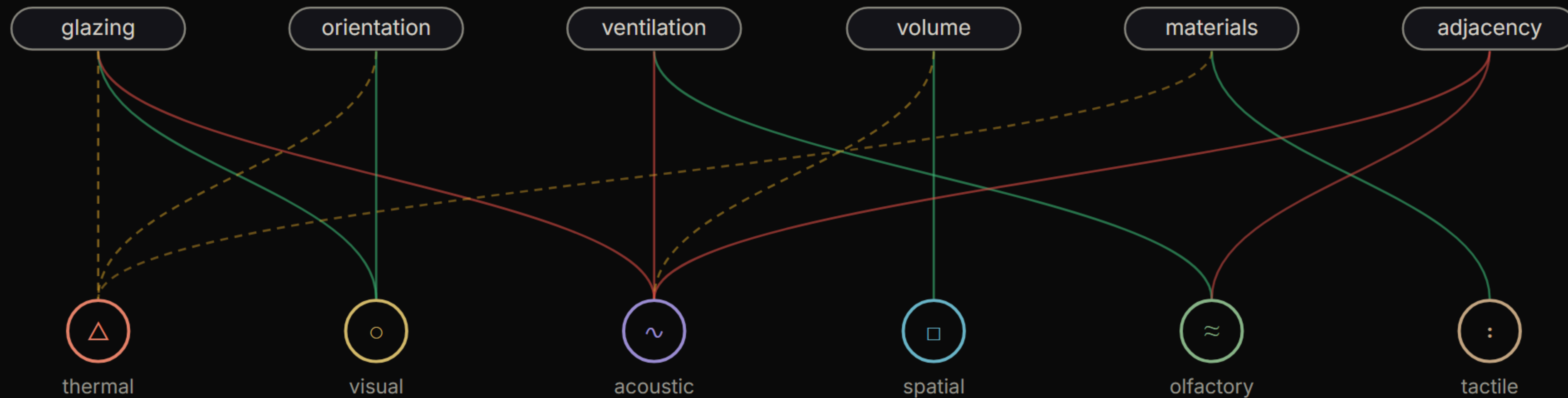


the card on the left is what these inputs produce. persona compiled, now we shape a plan: Act 2.

THE DESIGN LEVERS

Six design moves shape every sense.

orientation, glazing, ventilation, volume, materials, what's next door, each pulls on one or more senses.



+ helps - harms ± trade-off

solid = research-verified dashed = room physics

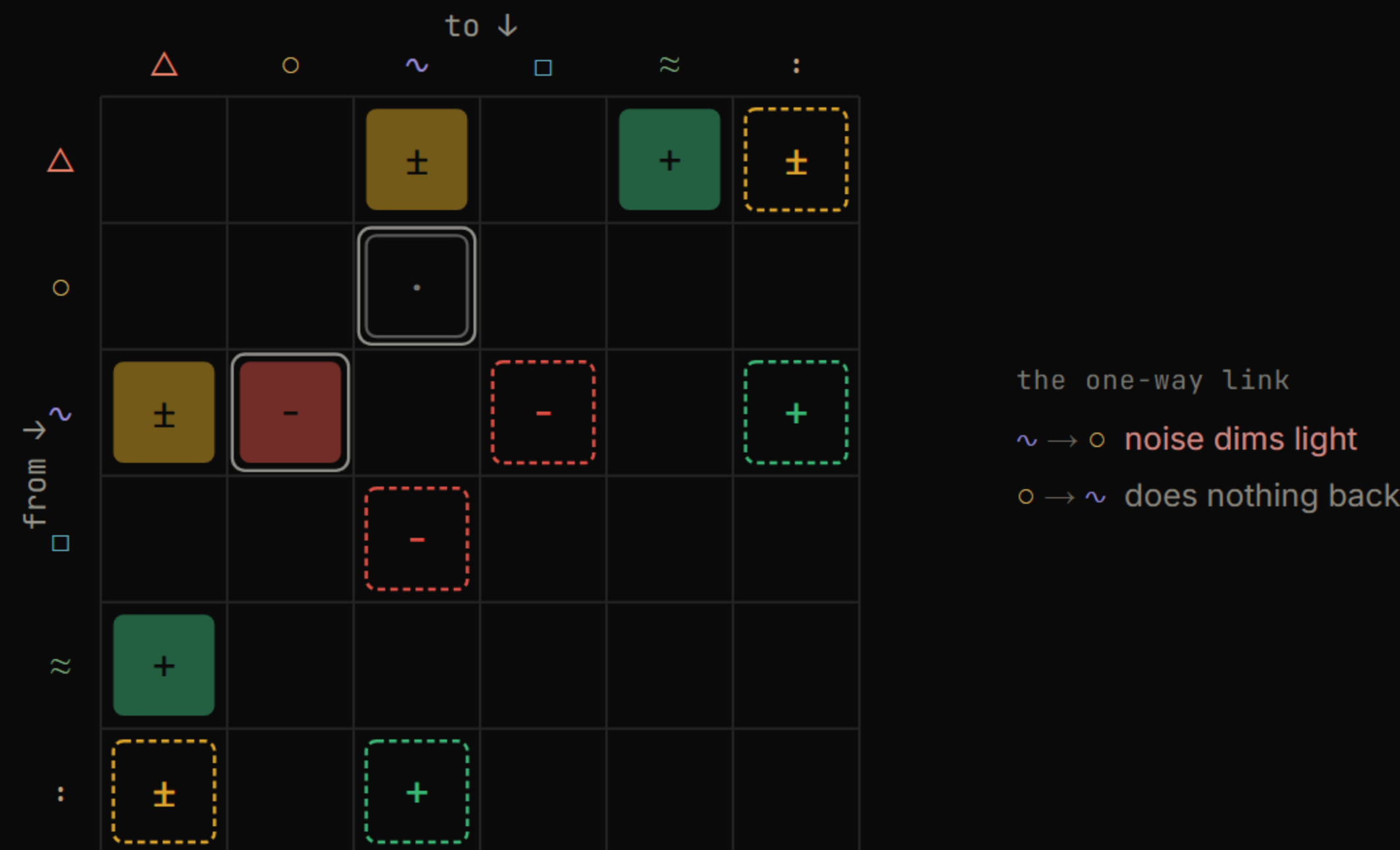
S5-S7 • standards BS 8233, ASHRAE 62.2, daylight metrics

R6 • materials Velt et al., 2012, tactile warmth from the surface

THE RELATIONSHIP BETWEEN SENSES

Each sense moves the others.

each link is research-verified or physics-based, and 18 of 25 survived a 3-vote check; then the worst sense vetoes the average.



THEN THE VETO no averaging, the worst sense floors the score: $\frac{1}{2}$ mean + $\frac{1}{2}$ worst → a kitchen fine except for smell **0.20** lands at **0.34**, not 0.49

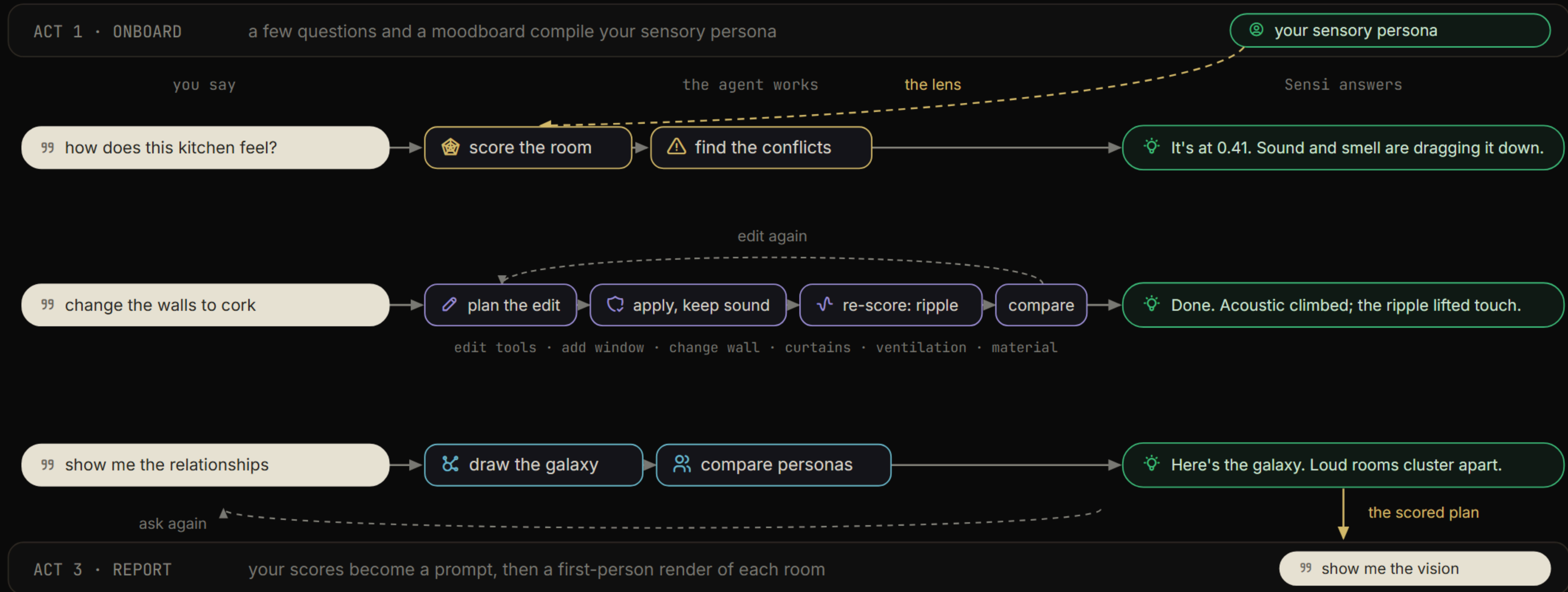
+ helps - harms ± trade-off · none
solid = research-verified dashed = room physics

18 / 25 couplings survived a 3-vote adversarial check
Yang & Moon, 2019 the acoustic-thermal link and the one-way acoustic-visual asymmetry

— ACT 2 / SHAPE • THE AGENT

Now you shape the plan, just by asking.

the agent at work: you change the plan in plain language, and one read routes each turn to the right capability.



three kinds of work • understand • edit, the ripple • see it anew • you • Sensi ... persona carries forward



SENSI

| shape |

• CITY APARTMENT •

0.40

E

PROFILE

We added a double East window in the Kitchen to increase glazing from **0.08** to **0.20**, and added a double South window in the Living Room. This modification shifted Living Room Δ thermal from **0.66** $\rightarrow$ **0.69** and $\approx$ olfactory from **0.73** $\rightarrow$ **0.77**, while Kitchen Δ thermal decreased from **0.19** $\rightarrow$ **0.17** and $\circ$ visual increased from **0.47** $\rightarrow$ **0.69**, with no change in the other rooms. As a next step, we should run a conflict analysis to ensure these glazing adjustments do not negatively impact $\sim$ acoustic performance.

Commit this draft to save the current layout, or run a full analysis to see how these changes impacted the $\sim$ acoustic performance in the Kitchen. You could also ask me why the Kitchen $\sim$ acoustic score remains so low, or just ask me anything about the results.

UNCOMMITTED CHANGES • SINCE THE LAST CHECKPOINT

$\circ$ 0.55 $\rightarrow$ 0.58 $\uparrow$

name this milestone (optional)

✓ commit checkpoint



ask sensi about your layout



< chat

plan

comfort

graph

galaxy ↗

green ↗

the vision →

• uncommitted edits not in The Vision yet

low $\rightarrow$ good $\circ$ room score Δ N



KITCHEN

0.24 ×



SENSES

Δ thermal 0.17 0.36 $\rightarrow$ 0.17 $\uparrow$ $\circ$ visual 0.69 0.78 $\rightarrow$ 0.69 $\uparrow$ $\sim$ acoustic 0.04 0.24 $\rightarrow$ 0.04 $\uparrow$ $\square$ spatial 0.87 1.00 $\rightarrow$ 0.87 $\uparrow$ $\approx$ olfactory 0.23 0.30 $\rightarrow$ 0.23 $\uparrow$ $:$ tactile 0.36 0.45 $\rightarrow$ 0.36 $\uparrow$

CONFLICTS - BELOW YOUR THRESHOLD

Δ thermal $\circ$ visual $\sim$ acoustic
 $\approx$ olfactory $:$ tactile

FIX IT - RUN A WHAT-IF

✗ glazing type ✗ glazing ratio
✗ plants ✗ surface material

SUGGESTIONS

✓ Add larger windows or skylights
 $\approx$ Upgrade to mixed or mechanical ventilation for active air exchange
 $\sim$ Add rugs and curtains to absorb sound
 $:$ Add natural material surfaces (wood, cork)

✗ Upgrade to double or triple glazing to

• CHECKPOINTS • INITIAL LAYOUT 0.30 $\uparrow$ 6

GRAPH

— SCORING ENGINE

How a room's six senses become one number.

layout 202: the same map, now lit on one room. every number traces back to the plan, no black box.



THE ENGINE deterministic, no LLM: **baseline** → **rules** → **couple** → **veto** → **0.41** a plain average would say 0.53

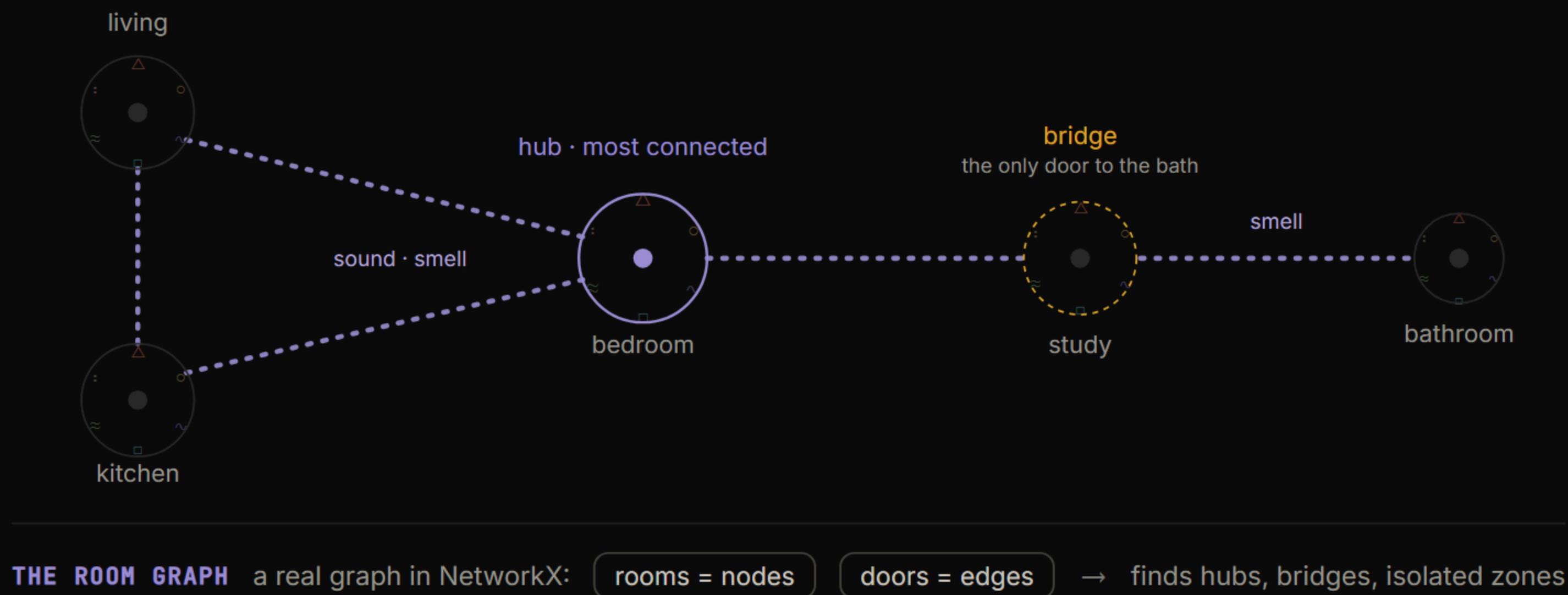
S5-S7 • **standards** BS 8233 acoustic, ASHRAE 62.2 ventilation, residential daylight

S2 • **the veto** Yang & Moon, 2019, non-additive aggregation **R6** material physics

ROOM TO ROOM

Rooms are wired together by their senses.

the kitchen's noise and smell reach the bedroom, the busiest room, and that is what zoning fixes.

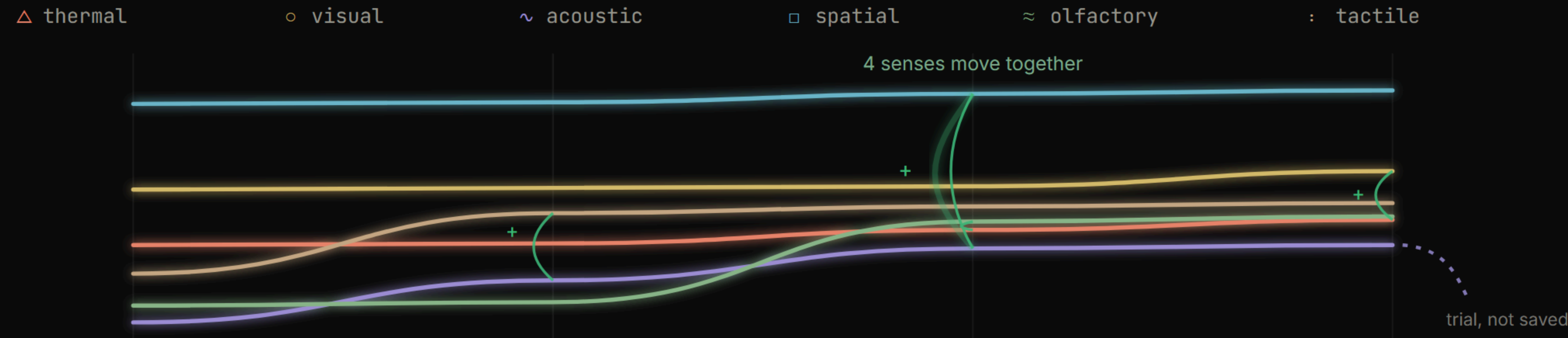


R1 · sensory zoning Al-Harasis et al., 2025, a taxonomy for sensory design in autism

— EVERY EDIT, TRACKED

Only the edits you commit reach the Report.

each line is a sense, rising as you edit; a layout-wide move ripples through everything, a single room far less.



CHECKPOINTS

● INITIAL

0.36

✓ + CORK room

0.40

✓ + OPEN THE PLAN layout

0.44

✓ + WINDOW room

0.46

○ + SKYLIGHT trying

→ Act 3

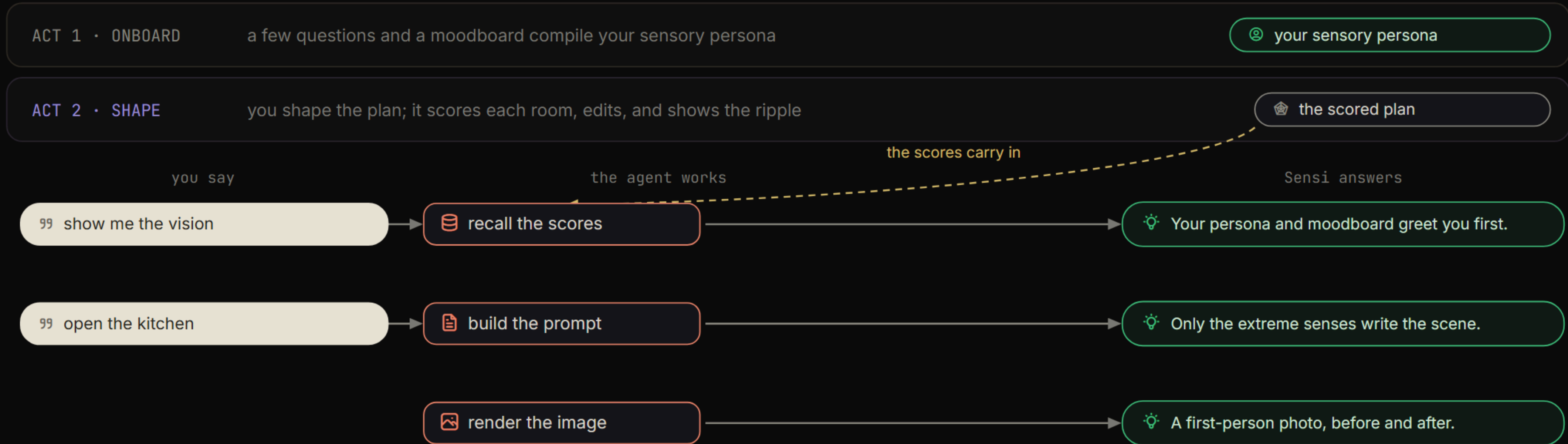
scope shows in the weave: a room edit moves a sense or two, a layout edit ripples through everything. the coupling, not noise.

only committed checkpoints carry forward: the dashed trial stays here, the Report renders the committed plan.

— ACT 3 / REPORT

Act three: the output answers the input.

the loop closes: your persona and aesthetic return, and the scores become a first-person image.



the output answers the input: the loop closes.



spatial

WHAT YOUR CHANGES DID - KITCHEN · FLOORMATERIAL, WALLMATERIAL, GLAZINGRATIO, FURNITURE · INITIAL → NOW

kitchen **0.38**whole home **0.47**

BEFORE · THE PROMPT

First-person, eye-level interior photograph of a kitchen. The floor is ceramic. The space feels a cool, slightly cold feel with bluish daylight; hard reflective surfaces – bare concrete, glass and tile – that look acoustically live; open, spacious and generous in volume; stuffy and closed with stale air; cold, hard, unwelcoming materials. Furnishings include a stone island, a ceramic counter, a metal sink, a metal fridge. Natural perspective, 35mm lens, photorealistic, high detail, no text, no people. Shot as restrained, material-honest architectural photography.

AFTER · THE PROMPT

First-person, eye-level interior photograph of a kitchen. The floor is wood. The space feels bright, airy and visually calm with balanced natural light; hard reflective surfaces – bare concrete, glass and tile – that look acoustically live; open, spacious and generous in volume. Furnishings include a stone island, a ceramic counter, a metal sink, a metal fridge, a fabric rug, three potted plants. Natural perspective, 35mm lens, photorealistic, high detail, no text, no people. Shot as restrained, material-honest architectural photography.

— SCORES → PROMPT → IMAGE

A score becomes a room you can stand in.

only the extreme senses speak, and the same room, edited, before and after.

HOW THE PICTURE IS WRITTEN

the scores

becomes a prompt



First-person photo of a kitchen.

hard, echoing surfaces,

open, generous in volume,

stale, stuffy air,

cold, hard materials.

Even, neutral light.

acoustic ↓ spatial ↑

olfactory ↓ tactile ↓

thermal and visual stay silent

render



THE SAME ROOM, BEFORE AND AFTER



wipe between them: same room, same angle, only the change moves.

the look you curated at minute one returns here. input and output meet.

— WHAT WE LEARNED

The wins, and the questions they raise.

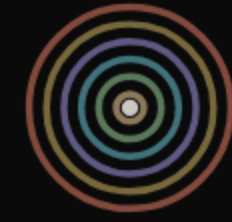
✓ WHAT HOLDS

-  **ripple and veto reprioritised the rooms**
we worried about different rooms than the headline number
-  **routing ~0.6s, the felt read ~24s**
routing only labels the turn; the felt read reasons over all six scores, their couplings, and your persona
-  **renders 2.75x faster than OpenAI**
\$0.039 vs \$0.042 a room; about 4x cheaper than OpenAI's premium tier

⚠ WHAT STAYS OPEN

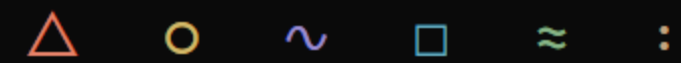
-  **Could we ever prove it?**
no ground-truth data to check it against; 18 of 25 couplings survived a model vote: agreement, not truth
-  **What if a sense doesn't apply?**
for a blind or Deaf occupant, could Sensi drop that channel and rebalance its weight onto the senses they live by, so the space works harder through those?
-  **Help, or decide for you?**
a confident score is easy to just follow; does Sensi help a designer decide, or quietly decide for them?

≡ We made the layer of the senses legible. What else can we make legible?



thank you

comfort, made legible while a building is still a plan



Charles Abi Chahine · Emilie El Chidiac · Lakzhmy Zaro · María Sánchez Domínguez